



Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide – QT114A

Stripped or damaged threads in engine component bolt holes may be repairable using the TimeSert® Thread Repair Kit. This Quick Training Guide describes how to install the TimeSert® thread repair insert.



Click the ENTER button to view the Quick Training Guide.

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

Overview

TimeSert® Kit

Inspection

Preparation

Setup

Drilling

Tapping

Installing

Overview

 Click the tabs at the top to view each section

When the threads in an engine component bolt hole become stripped or damaged, it may be possible to repair the bolt hole using a thread insert. The repair process involves drilling out the bolt hole to a slightly larger diameter, tapping the hole, and then installing an appropriately-sized thread repair insert in the hole.

The basic installation steps are:

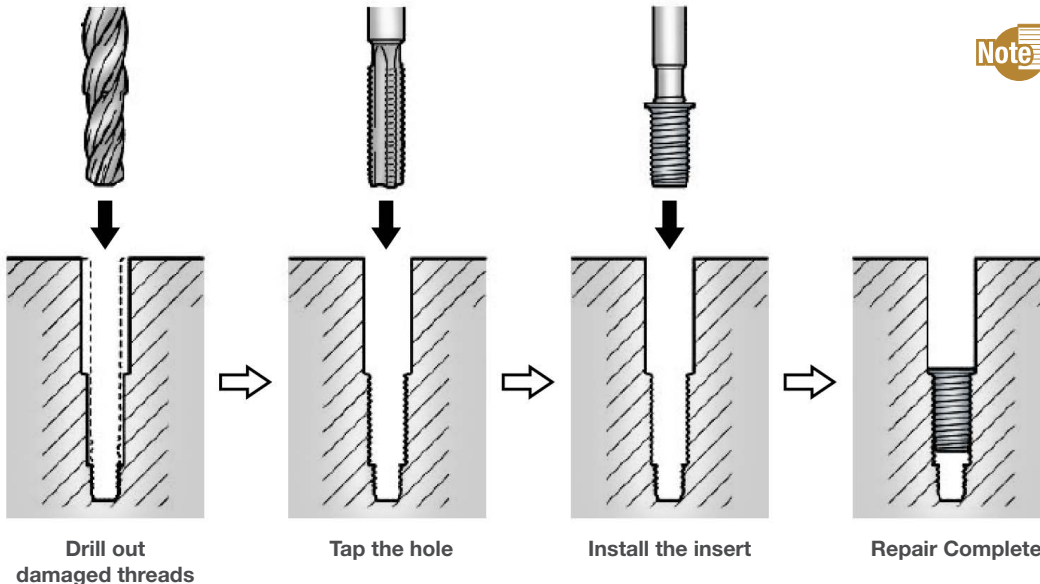
- Inspecting the components and bolts for serviceability
- Fastening the drill fixture to the component (when applicable)
- Drilling out the damaged bolt hole
- Tapping the bolt hole
- Installing the insert

Precautions

- Protective goggles, a protective mask, and other necessary protective equipment must be worn while working
- Brake cleaner is highly flammable; do not use it near fire or hot components
 - If you get brake cleaner on your clothes, keep away from fire and hot components until clothes are dry
 - Be careful to prevent spattering while working
 - For details, refer to the brake cleaner's handling instructions



Before installing a TimeSert® kit in a Toyota engine, check for applicable Technical Service Bulletins.



Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

[Overview](#)[TimeSert® Kit](#)[Inspection](#)[Preparation](#)[Setup](#)[Drilling](#)[Tapping](#)[Installing](#)

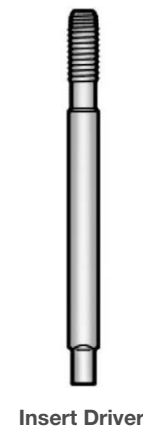
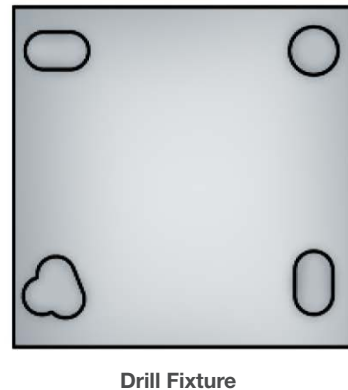
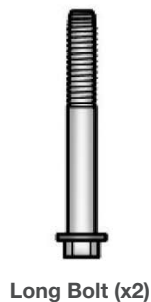
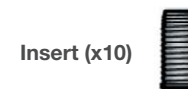
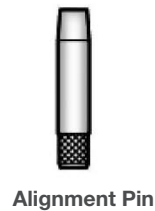
The TimeSert® Kit

TimeSert® kits and components may be ordered by calling Approved Dealer Equipment (ADE) at 1-800-368-6787.

When ordering, be sure to specify:

- The correct size (for example, M11 x 1.5 x 30mm)
- The number of inserts needed

 Click the tabs at the top to view each section



Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

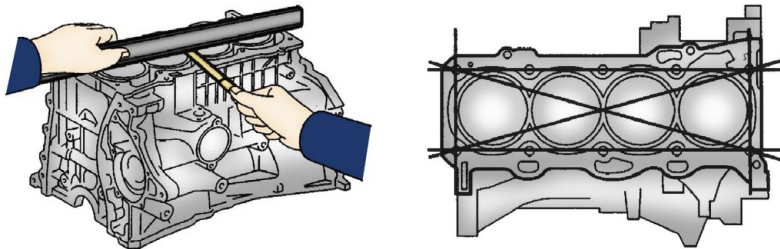
[Overview](#)[TimeSert® Kit](#)[Inspection](#)[Preparation](#)[Setup](#)[Drilling](#)[Tapping](#)[Installing](#)

Inspection

With the engine removed from the vehicle and mounted on an engine stand, disassemble the engine to gain access to the necessary components. Perform the following inspections:

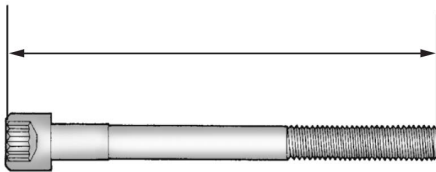
- Measure the component mating surfaces for flatness
- Measure the bolts for stretch

Example: Measuring the Cylinder Block for Flatness



Using a precision straightedge and feeler gauge, measure the cylinder head contact surface for flatness. If the Warpage exceeds Repair Manual specifications, replace the short block assembly instead of attempting to repair it.

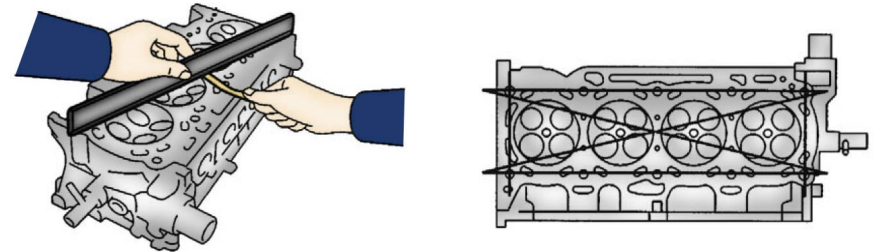
Example: Measuring Cylinder Head Bolts



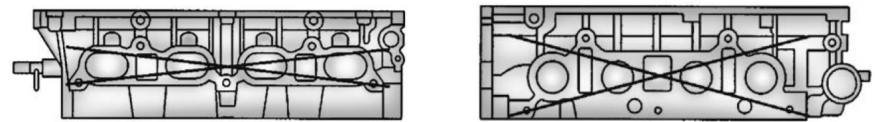
Using a caliper, measure the length of the head bolts from the seat to the end. If any bolt exceeds Repair Manual specifications, replace the bolt.

 Click the tabs at the top to view each section

Example: Measuring the Cylinder Head for Flatness



Using a precision straightedge and feeler gauge, measure the cylinder head contact surface for flatness.



Also measure the intake and exhaust manifold contact surfaces for flatness.

If the warpage exceeds Repair Manual specifications, replace the component instead of attempting to repair it.

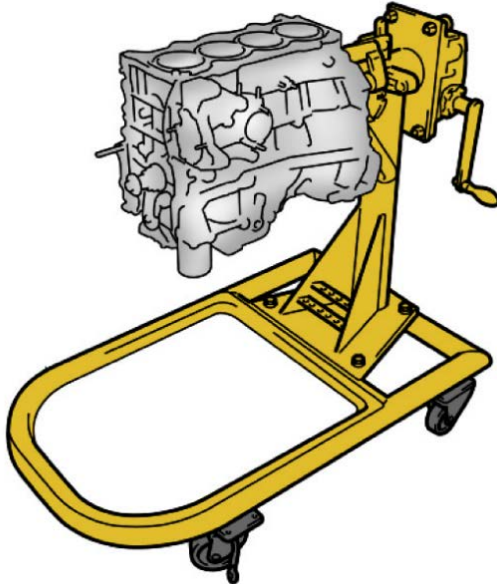
Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

[Overview](#)[TimeSert® Kit](#)[Inspection](#)[Preparation](#)[Setup](#)[Drilling](#)[Tapping](#)[Installing](#)

Preparation

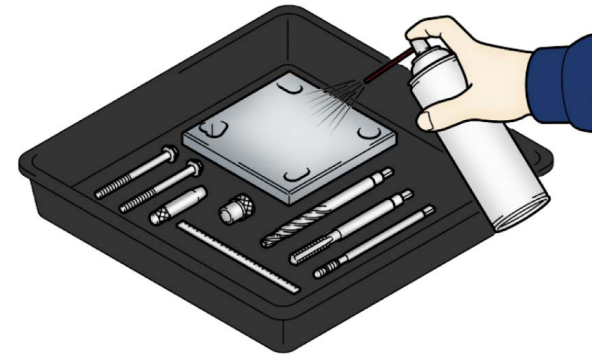
Attach the engine or engine component to the engine stand and make sure that it can be rotated.



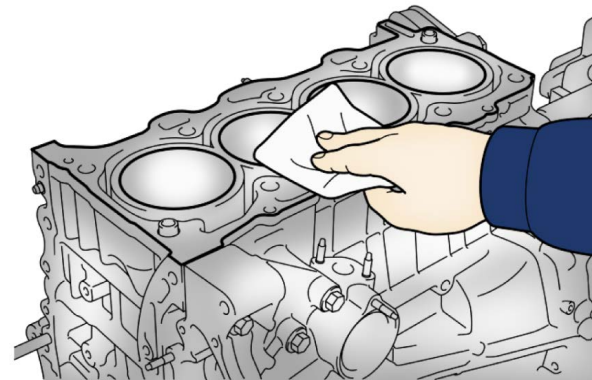
 Click the tabs at the top to view each section

Before beginning work:

- Clean off any cutting particles or other substances from the tools that will be used



- Remove any oil or other substances from the component surface



Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

Overview

TimeSert® Kit

Inspection

Preparation

Setup

Drilling

Tapping

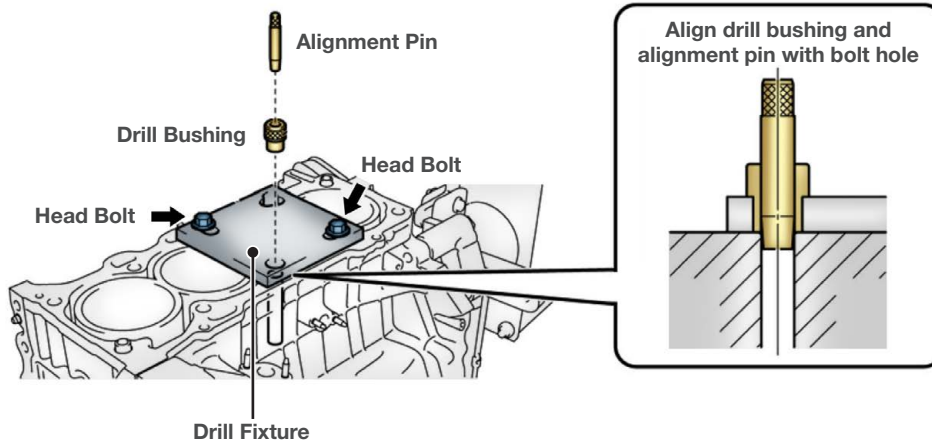
Installing

Setup

1. Align the Drill Fixture

When applicable, attach the drill fixture to the engine component you're working on.

- Align the round hole of the drill fixture with the bolt hole which will be repaired.
- Install the drill bushing and alignment pin, and then temporarily install 2 bolts to fix the drill fixture.

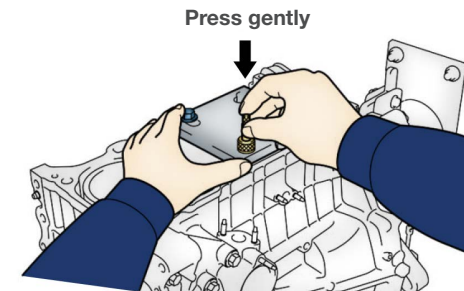


Note When repairing bolt holes in the cylinder block:

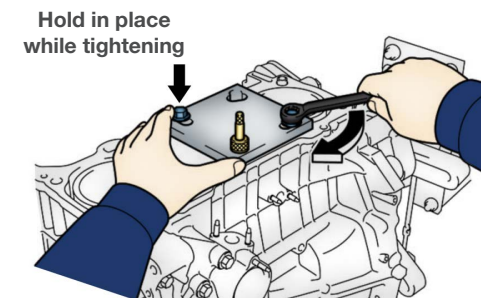
- Move the pistons from the Top Dead Center position in order to avoid contact between pistons and drill fixture
- Always use 2 bolts to fix the drill fixture
- Avoid using bolt holes with damaged threads for this procedure and install the bolts diagonally whenever possible

2. Fasten the Drill Fixture

- Press the alignment pin gently and make sure that the pin is inserted all the way down into the hole.



- Hold the drill fixture so that it does not become displaced, and tighten the 2 bolts.



- Remove the alignment pin, and visually check that the bolt hole and drill fixture are centered.



CAUTION: If the drill fixture is NOT correctly centered, the bolt holes may be drilled at an angle, and the component will need to be replaced.



Click the tabs at the top to view each section

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

[Overview](#)[TimeSert® Kit](#)[Inspection](#)[Preparation](#)[Setup](#)[Drilling](#)[Tapping](#)[Installing](#)

Drilling: Protect the Component

 Click the tabs at the top and side to view each section

If the cutting particles that are produced during the drilling and tapping process enter the engine, they can cause engine damage, so be sure to apply protection.

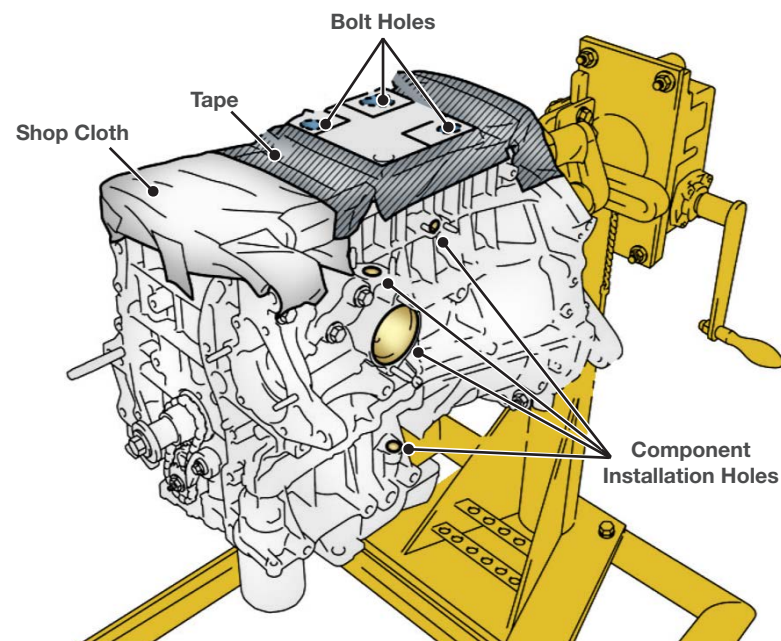
Example:

When repairing the cylinder block, use tape or a shop cloth to cover the parts shown and protect the cylinder block so that cutting particles do NOT enter:

- Drill fixture bolt holes
- Top surface of the cylinder block
- Cylinder block component installation holes



Use a new shop cloth and tape for each repair as the old ones may have cutting particles attached to them.



Protect the Component

Drill the Bolt Hole

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

Overview

TimeSert® Kit

Inspection

Preparation

Setup

Drilling

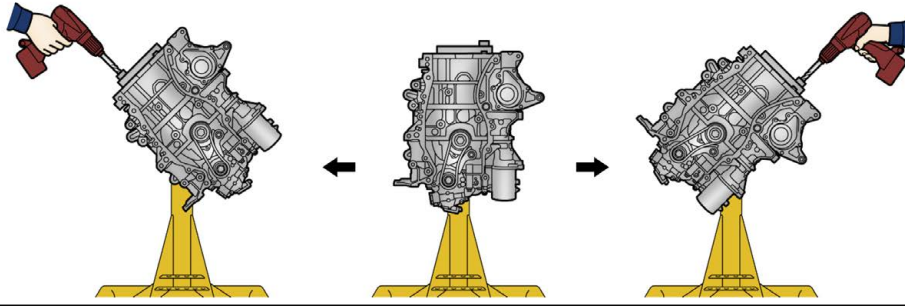
Tapping

Installing

Drilling: Drill the Bolt Hole

 Click the tabs at the top and side to view each section

1. Tilt the component being repaired so that the cutting particles will fall away from the component.

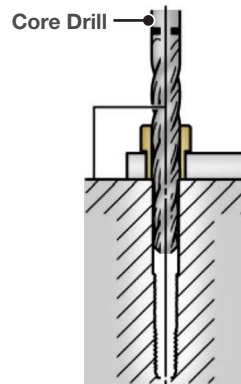
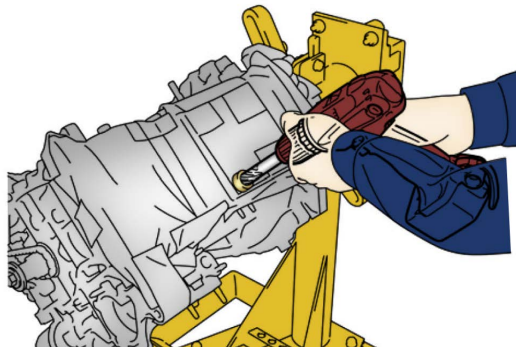


2. Use the core drill and drill the bolt hole.



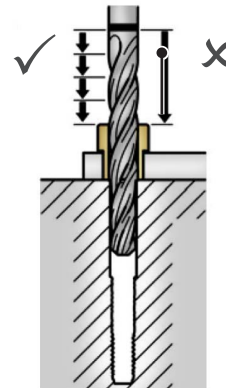
CAUTIONS:

- Hold the drill perpendicular to the top surface of the component. If the bolt holes are drilled at an angle, the component will need to be replaced.
- Drill the hole in **4 progressively deeper steps**, retracting the drill each time, to clear the cutting particles from the hole.
- Apply WD-40® regularly to lubricate the drill.

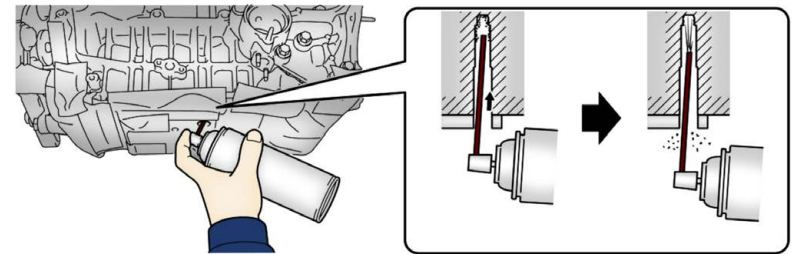


Drill perpendicular
to mating surface

Drill in 4 steps. Do not
drill in one step.



3. Remove the core drill and drill bushing. Invert the component so that it is upside-down. Insert the nozzle all the way into the bolt hole and spray brake cleaner into the hole in order to flush out the cutting particles from inside it.



4. Return the component to a tilted position and continue drilling in steps until you reach the bottom of the hole.

Protect the Component

Drill the Bolt Hole

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

Overview

TimeSert® Kit

Inspection

Preparation

Setup

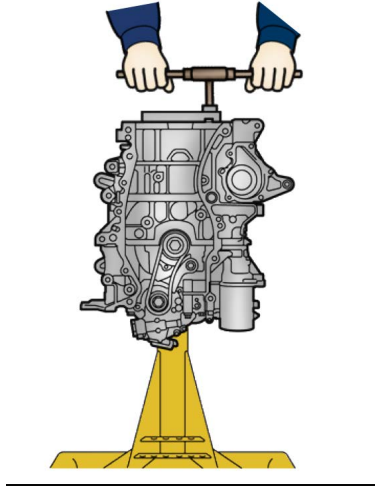
Drilling

Tapping

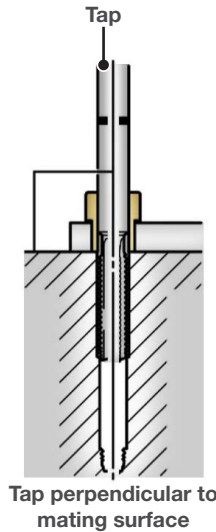
Installing

Tapping: Using the Tap

1. Position the component vertically, and use the tap and tap handle to cut new threads in the bolt hole.

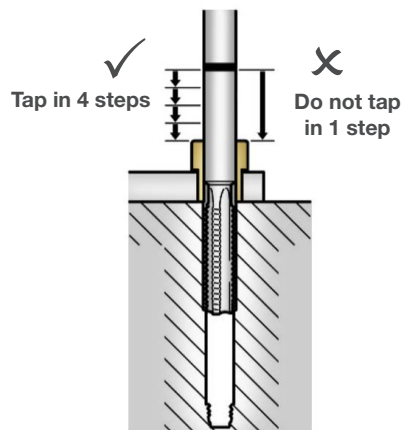


 Click the tabs at the top and side to view each section

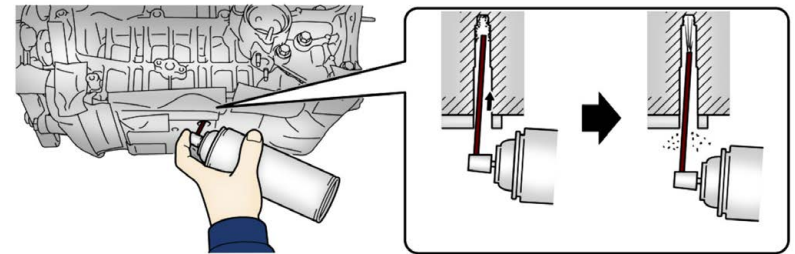


CAUTION: Hold the tap perpendicular to the top surface of the component. If the bolt holes are tapped at an angle, the component will need to be replaced.

- Tap the hole in **4 progressively deeper steps**, retracting the tap each time, to clear the cutting particles from the hole.
- Apply WD-40® regularly to lubricate the tap.



2. Invert the component so that it is upside-down. Insert the nozzle all the way into the bolt hole and spray brake cleaner into the hole in order to flush out the cutting particles from inside it.



3. Return the component to a tilted position and continue tapping in steps until you reach the bottom of the hole.

Using the Tap

Drying the Bolt Hole & Checking for Leaks

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

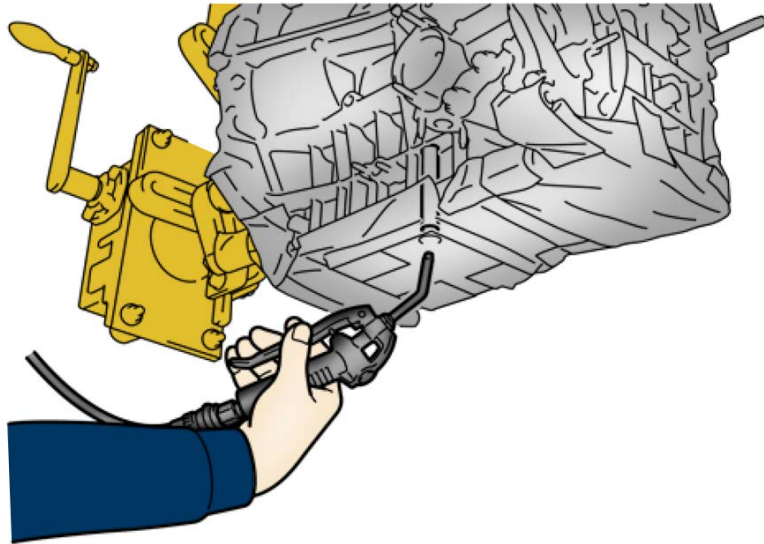
[Overview](#)[TimeSert® Kit](#)[Inspection](#)[Preparation](#)[Setup](#)[Drilling](#)[Tapping](#)[Installing](#)

Tapping: Drying the Bolt Hole and Checking for Leaks

 Click the tabs at the top and side to view each section

Drying the Bolt Hole

After the final tap, invert the component and clean the bolt hole. Use an air blower and dry the hole completely.



Because Loctite® will be used to fasten the insert in place, be sure to dry the bolt hole thoroughly while the component is inverted.

Checking for Leaks

Use a cylinder leak down tester and check the bolt holes.

- Wrap the tip of the tester with seal tape so it is thick enough to fill the gap between the tip of the tester and the hole
- Apply 44 psi (300 kPa) to the bolt hole
- Confirm the air pressure does NOT decrease after 15 seconds



- When repairing a cylinder block, if air leaks through the bolt hole into the cylinder block, replace the short block
- Make sure to perform the leak check BEFORE inserting the insert to the bolt holes

Using the Tap

Drying the Bolt Hole & Checking for Leaks

Toyota Thread Repair Using TimeSert® Thread Repair Kit

Quick Training Guide QT114A

Overview

TimeSert® Kit

Inspection

Preparation

Setup

Drilling

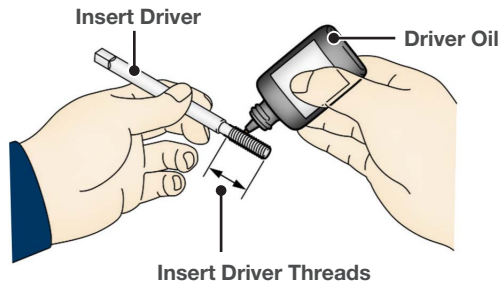
Tapping

Installing

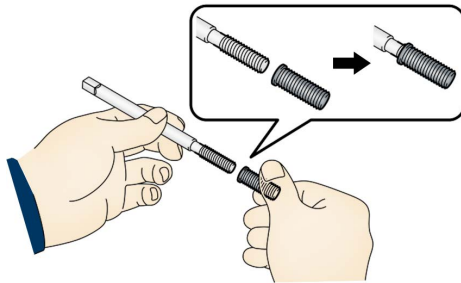
Installing

 Click the tabs at the top to view each section

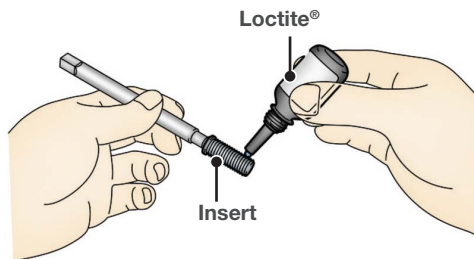
1. Apply a small amount of driver oil to the threads of the insert driver.



2. Screw the insert onto the insert driver until it stops. Do not apply oil to the outer threads of the insert.

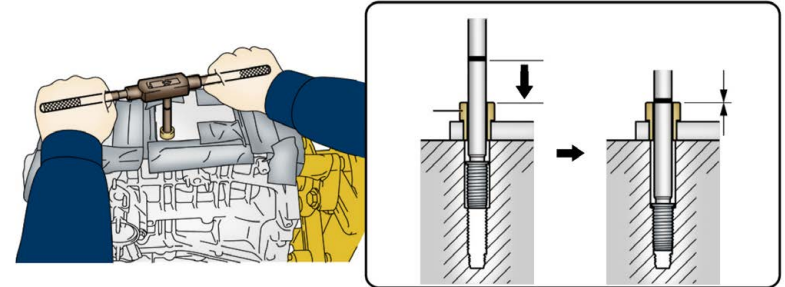


3. Apply a small amount of Loctite® to the outer threads of the insert.



Note The insert will not screw all the way onto the driver because the last few threads of the insert are unfinished. Installing the insert will finish the last 3 threads and lock the insert in place.

4. Insert the insert driver with the insert into the bolt hole and slowly screw it into the hole.



Note Near the bottom of the hole, you will begin to feel resistance as the driver expands the last 3 threads to lock the insert in place. When the driver begins turning easily again, the installation is complete. Turn the driver in the opposite direction and remove it.

Test the Repair

Remove the drill bushing from the fixture. Apply a small amount of engine oil to the threads of the bolt, and screw the bolt into the hole by hand to verify that there is no problem with the threads.

Remove the shop cloth and protective tape, and remove the drill fixture (when applicable).

Note Work carefully so the cutting particles do not fall on the engine.